DNS roadmap notes

Session record, 2026-07-17 — Re 800 validation, solver directions, compressible pivot

1 Current state (validated)

Incompressible DNS: MAC grid, RK3 (SSP) + pressure projection, KE-conserving skew-symmetric advection — the only dissipation is the physical ν. HIP GPU backend (FP32 fields, FP64 reductions) mirrors the CPU reference 1:1.

- **Sharp viscous wall** (ghost fill, `src/dns/gpu/ghostfill.*`): sub-cell no-slip via mirror ghosts fed *only* to the diffusion term (ghosts in the skew convection inject energy, measured +2.3%/300 inviscid steps). Grid-converged: cylinder Re 60 $c_d = 1.785$ at both 512^2 and 1024^2 the old staircase wall sat 0.066 low.
- **Free-slip lateral tunnel walls** (all four tunnel apps): tangential mirrored (wall vorticity $\equiv 0$, verified $5 \cdot 10^{-4}$ max), normal antimirrored, pressure Neumann. The earlier Dirichlet $u = u_\infty$ clamp painted a real vortex sheet (wall $|\omega|$ 1.2–3.6 vs 0.016 interior) — do not reintroduce.
- **Latest run** — circular cylinder, $\mathrm{Re}_D = 800$, 512^2, 15 flow-throughs, pure DNS: $c_d = 1.832 \pm 0.218$, $\mathrm{St} = 0.257$, c_l rms 1.17 (peaks ± 1.6). Honesty: at Re 800 the *2D restriction itself* is the dominant model error (real wake is 3D past Re ≈ 190; experiment $c_d \approx 1.0$, 2D characteristically 1.8–2.0). Boundary layer $\approx D/\sqrt{\mathrm{Re}} \approx 2.3$ cells — just under the 3-cell rule.

2 Grounding against reality and literature

Cheapest-first ladder:

1. **Canonical validation suite as ctest targets.** Cylinder Re 100 (c_d 1.33, St 0.164), Re 200 (St 0.197), sphere Re 300 (c_d 0.657, c_l 0.069 Johnson & Patel — we measure 0.706/0.071), 3D Taylor–Green Re 1600 dissipation curve vs the spectral reference.
2. **Blockage correction.** The $8D$ tunnel over-reads c_d by 5–10% (Maskell); likely most of the sphere 0.706-vs-0.657 gap. Validate at 16–24D or report corrected values.
3. **Convergence triples** (256/512/1024) with Richardson extrapolation; report the extrapolated value and the observed order, always.
4. **Discrete KE budget closure** per run: $\mathrm{d}K/\mathrm{d}t + \varepsilon_\nu + \text{body drain} - \text{boundary flux} = 0$ to roundoff. Would have caught the Dirichlet wall sheet immediately.
5. FP64-field anchor run to bound FP32 drift; 3D spectra ($-5/3$ band) and enstrophy vs published TGV data.

3 Cylindrical coordinates — verdict: skip

A native (r, θ, z) MAC grid costs: metric terms that break the exact skew-form KE conservation, the $r = 0$ axis singularity, Δt collapse from tiny $r\Delta\theta$ cells, and a second Poisson family (FFT in θ, z + tridiagonal in r). Everything cylindrical geometry buys — pipes, annuli, rotating and axisymmetric bodies — the validated cut-cell + ghost-fill machinery already handles on the Cartesian grid. Only worth revisiting for rotating-machinery flows where azimuthal periodicity buys spectral accuracy in θ.

4 Arbitrary geometry: STL / glTF / GLB

Highest value-per-effort item. The cut-cell + ghost machinery needs exactly two callables — $\mathrm{sdf}(x, y, z)$ and cut geometry (λ, apertures) — so mesh support is a *loader*, not a solver change:

1. Triangle soup from STL (trivial) or glTF/GLB via `cgltf`.
2. Signed distance: BVH + closest triangle, sign from angle-weighted pseudonormals (Bærentzen–Aanæs) or generalized winding number (Jacobson) for leaky scans.

3. Precompute a narrow-band SDF voxel grid (2× flow resolution) once on CPU, upload as a 3D texture; ghost fill and the existing supersampled cut classification (`makeGolfCut` pattern: 8^3 samples per cell) consume it unchanged.
4. Deliverable: `mesh_tunnel_gpu model.stl --scale --pitch --yaw` — one new app, zero kernel changes.

5 Body-fitted O-grids (the "X-wing" idea)

Wrap the body outline with the ξ index ($\eta = 0$ *is* the wall), march η outward.

2D special case is genuinely attractive. A polygonal cross-section (the + shape) admits a Schwarz–Christoffel conformal map from an annulus to its exterior. Conformal ⇒ scalar metric $J = |f'|^2$, exactly orthogonal grid, and the transformed pressure Poisson stays *constant-coefficient*: $\partial_{\xi\xi} p + \partial_{\eta\eta} p = Jr$ — FFT in periodic ξ

1. tridiagonal in η. The fast exact solver survives. Contravariant volume fluxes on a

staggered computational grid keep discrete continuity and exact skew-form KE conservation (Zang et al. 1994). Wall BC is a grid line: no cut cells, no ghost fill, and wall-normal clustering ($\Delta y^+ \approx 1$) is free.

The catch: corners. Conformal maps are singular at polygon corners — $|f'| \to 0$ at convex corners (Δt collapse), $\to \infty$ at the four reentrant corners (resolution starves exactly where the shear layer separates). Fix: round the corners with radius of a few cells (physical anyway). Non-conformal alternative (Winslow smoothing) improves grid quality but makes the metric a tensor: variable-coefficient Poisson, CG/multigrid only.

3D reality. A single structured O-grid around a full aircraft does not exist topologically; industry uses multi-block/overset (Chimera, e.g. OVERFLOW) — a large lift (hole cutting, fringe interpolation, approximate conservation across fringes). Not planned.

Recommended order: STL support → co-moving frame + time-dependent SDF (pitching bodies, nearly free) → 2D conformal O-grid mini-solver as a validation cross-check → forget native cylindrical.

6 Pressure solver: one-shot vs preconditioned, and the iteration ladder

Without a body the projection *is* one-shot: FFT/DCT → divide by discrete-FD Laplacian eigenvalues → inverse transform, machine-exact. With cut cells the operator becomes $\mathrm{div}(a \cdot \mathrm{grad}\ p)$ with per-face apertures $a \in [0, 1]$ near the surface — not translation-invariant, so no diagonal multiply exists. The uncut FFT inverse is instead used as a *preconditioner* inside CG; the perturbation lives on a thin shell, so iterations stay ≈ 40 independent of grid size.

Reduction ladder (pressure dominates step cost: 3 stages × 40 iters × one FFT round-trip each):

1. **Extrapolated warm start** (trivial): $\varphi_{\text{guess}} = 2\varphi_{n-1} - \varphi_{n-2}$, kept per RK stage. Expected 20–40% fewer iterations. (Plain warm start is already in.)
2. **Tolerance audit** (free): FP32 advection noise re-injects divergence next step; find where the useful residual decade ends.
3. **Deflated PCG** (moderate): the slow modes live on the static cut shell — precompute them once (Lanczos at startup), deflate. 40 → 10–15. Works in 2D and 3D.
4. **Capacitance matrix / Woodbury** (2D kill shot): $A = \hat{A} + P$ with P of rank $m \approx$ #cut faces (800 for the 512^2 cylinder). Factor the $m \times m$ capacitance matrix once (Proskurowski–Widlund); every projection becomes 2 FFT solves + one small triangular solve — **exact, zero iterations**. In 3D, $m \sim 30$k surface faces makes the dense factor 7 GB: fall back to deflation or an $\mathcal{H}$-matrix.

Expected: 3–4× faster 2D steps, 2–3× in 3D.

7 Compressible Navier–Stokes — next step

Motivation: acoustics, genuine Mach effects, and a solver with *no elliptic solve at all* (fully explicit — embarrassingly GPU-friendly; the trade is the acoustic CFL $\Delta t \propto h/(|u| + c)$).

Plan (house pattern: CPU double-precision reference first, GPU port after validation):

- Conservative variables $q = (\rho, \rho u, \rho v, E)$, ideal gas $\gamma = 1.4$, $p = (\gamma - 1)\big(E - \frac{1}{2}\rho\,|u|^2\big)$. Collocated grid (no MAC needed — no projection).
- Convective terms: Kennedy–Gruber kinetic-energy-consistent split-form two-point fluxes (the compressible analog of our skew form; central, non-dissipative, stable for smooth flows).
- Viscous terms: face-based compact stencils, Newtonian with Stokes hypothesis ($\lambda = -\frac{2}{3}\mu$), Fourier heat flux $k = \mu c_p/\,\mathrm{Pr}$, $\mathrm{Pr} = 0.72$.
- Time: SSP-RK3, acoustic CFL.
- **Stated limitation:** central non-dissipative fluxes — smooth/shock-free flows only. Shock capturing (HLLC/WENO hybrid or artificial bulk viscosity) is a later, separate step.
- Validation tests (`test_comp2d`): isentropic vortex advection (Euler; exact solution, measured order ≈ 2), small-amplitude acoustic pulse speed vs $c = \sqrt{\gamma p/\rho}$, low-Mach 2D Taylor–Green KE decay vs the incompressible exact rate $e^{-4\nu t}$, and exact conservation of $\sum \rho$, $\sum \rho u$, $\sum E$ to roundoff.